ASSIGNMENT : Interactive Dashboard & Tableau Storytelling

Dataset Link : Hostpital_Management

Students should submit:

Tableau Workbook file (.twb or .twbx)

Steps for each output (brief explanation)

Screenshots of Final Story Points

Short explanation of insights derived from each visualization

Problem Scenario

You are working as a Healthcare Data Analyst.

(A hospital management team wants to analyze patient trends, treatment cost, department performance, and regional insights using Tableau Dashboards and Story Points.

Your task is to build interactive visuals and present insights through storytelling. (Assignment Questions)

Q1 Explain the difference between a Tableau Dashboard and a Tableau Story Point. Give two healthcare-related use cases.

Sol: Difference between a tableau dashbbord and a tableau story point ;

TABLEAU DASHBOARD;

It combines multiple worksheets into single interactive screen.

It is used for monitoring KPISand Analysis.

User interacts with flter and charts.

TABLEAU STORYPOINT ;

It presents data as sequence of slides/pages for storytelling.

Used for presenting insights step by step.

User navigates through story points to understand finding.

Healthcare Use Cases;

Dashboard Use Cases

1. Monitor patient admissions, treatment costs, and department performance.

2. Track doctor performance and patient outcomes in real time.

Story Point Use Cases

1. Present patient trends across regions.
2. Explain treatment cost changes over time and departmental insights.

Q2 What are Actions in Tableau? Explain how Filter Action can help hospital management analyze patient data.

Sol: Actions allow users to interact with visualizations and affect other views.

Types of Actions;

- Filter Action
- Highlight Action
- URL Action

Filter Action in Healthcare;

Hospital management can click on a region in a bar chart and automatically view:

- Patients from that region
- Department performance
- Treatment costs
- Patient outcomes

This helps management quickly analyze specific patient groups without creating multiple reports.

Q3 Patient Distribution Create a Bar Chart showing: Total Patients by Hospital Region. Add this as a Story Point with proper title.

Sol: Create a Bar Chart showing Total Patients by Hospital Region.

Steps

1. Open Tableau.
2. Connect Hospital Management dataset.
3. Drag **Region** to Rows.
4. Drag **PatientID** to Columns.
5. Change aggregation to **COUNT(PatientID)**.
6. Select Bar Chart.
7. Sort descending.

8. Show data labels.
9. Title:

Patient Distribution by Hospital Region

Story Point 1

Title: Regional Patient Distribution

Insight

- Region with highest patients indicates higher healthcare demand.
- Region with lowest patients may require awareness or infrastructure improvements.

Screenshot Layout

```
------------------------------------------------
 Patient Distribution by Hospital Region
------------------------------------------------

West      ███████████████
East      ████████████
North     ██████████
South     ███████

-------------------------------------------
```

Q4 Department Performance Create a visualization showing: Treatment Cost by Department. Add labels and customize colors.

Sol: Treatment Cost by Department

Steps

1. Drag **Department** to Rows.
2. Drag **TreatmentCost** to Columns.
3. Select Bar Chart.
4. Drag TreatmentCost to Label.
5. Drag Department to Color.
6. Format currency.

Title

Treatment Cost by Department

Story Point 2

Department Performance Analysis

Insight

- Department with highest treatment cost consumes maximum resources.
- Helps management optimize budgeting.

Screenshot Layout:

```
------------------------------------------------
 Treatment Cost by Department
------------------------------------------------

Cardiology      █████████████
Neurology       ███████████
Orthopedics     ████████
Pediatrics      ██████

------------------------------------------------
```

Q5 Treatment Trend Over Time Create a Line Chart: Treatment Cost vs Admission Date. Filter the data only from March to September.

Sol : Line Chart: Treatment Cost vs Admission Date

Steps :

1. Drag AdmissionDate to Columns.
2. Change to Month.
3. Drag TreatmentCost to Rows.
4. Select Line Chart.
5. Add AdmissionDate filter.
6. Keep months:

- March
- April
- May
- June
- July
- August
- September

7. Show trend line.

Title:

Treatment Cost Trend (March–September)

Story Point 3

Treatment Cost Trend Analysis

Insight :

- Peak months indicate higher hospital expenditure.
- Helps forecast future budgets.

Screenshot Layout :

```
Treatment Cost

|
|                 *
|              *     *
|          *
|      *
|   *
----------------------------------------
 Mar Apr May Jun Jul Aug Sep
```

Q6 Outcome Analysis Create a Pie Chart: Patient Outcome Distribution (Recovered / Critical / Under Treatment).

Sol : Pie Chart: Patient Outcome Distribution

Steps :

1. Create a new sheet.
2. Drag Outcome to Color.
3. Drag PatientID to Angle.
4. Change Marks to Pie.
5. Add percentage labels.

Title :

Patient Outcome Distribution

Story Point 4:

Patient Outcome Analysis

Insight :

- High recovery percentage indicates effective treatment.
- Large critical cases may require additional resources.

Screenshot Layout :

```
        Recovered
           60%

     ◔◔◔◔◔◔◔◔◔◔

 Critical 25%
 Under Treatment 15%
```

Q7 Doctor Performance & Patient KPI Analysis

Task Part A

. Doctor Performance Create a Ranked Bar Chart showing:

. Top Doctors by Total Treatment Cost handled

Task Part B KPI Creation

. Create a KPI visualization showing:

. Total Patients Treated

Sol : TASK PART A:

Ranked Bar Chart: Top Doctors by Total Treatment Cost Handled

Steps

1. Drag DoctorName to Rows.
2. Drag TreatmentCost to Columns.
3. Sort descending.
4. Right-click DoctorName → Sort by SUM(TreatmentCost).
5. Add Rank:

Create Calculated Field:

```
RANK(SUM([TreatmentCost]))
```

6. Put Rank on Label.
7. Filter Top 10 Doctors.

Title :

Top 10 Doctors by Treatment Cost Handled

Story Point 5 :

Doctor Performance Analysis

Insight :

- Top doctors manage high-value or complex cases.
- Can be used for workload balancing and performance evaluation.

Screenshot Layout :

```
------------------------------------------------
 Top Doctors by Treatment Cost
------------------------------------------------

Dr A      ██████████████
Dr B      ████████████
Dr C      ██████████
Dr D      ████████
Dr E      ███████

------------------------------------------------
```

TASK Part B : KPI CREATION .

Create a KPI Visualization showing:

Total Patients Treated

Steps

1. Create a New Worksheet.
2. Drag **PatientID** to Text on the Marks Card.
3. Change aggregation to:

```
COUNT([PatientID])
```

4. Increase font size (24–36).
5. Format the KPI card.
6. Add title:

Total Patients Treated

Alternative Calculated Field :

Create a Calculated Field:

```
Total Patients Treated = COUNT([PatientID])
```

Place it on Text.

KPI Layout :

```
-------------------------------
       TOTAL PATIENTS
           5,000
-------------------------------
```

Insight :

- Shows the total number of patients handled by the hospital.
- Useful for monitoring hospital workload and capacity.

Q8 Interactive Dashboard

. Build a Dashboard containing:

. Region Chart

. Department Cost chart

. Outcome Chart

Add Filters:

. Hospital Region

. Department.

SOL : Create a Dashboard containing:

1. Region Chart
2. Department Cost Chart
3. Outcome Chart

Add Filters:

- Hospital Region
- Department

Dashboard Creation Steps:

Step 1: Create Dashboard;

1. Click **New Dashboard**.
2. Set Size → Automatic.

Step 2: Add Sheets;

Drag the following sheets into the dashboard:

- Patient Distribution by Region
- Treatment Cost by Department

- Outcome Analysis Pie Chart
- KPI Card (Total Patients Treated)

Step 3: Add Filters

Region Filter;

1. Drag Region to Filters.
2. Show Filter.
3. Apply to all worksheets.

Department Filter;

1. Drag Department to Filters.
2. Show Filter.
3. Apply to all worksheets.

Dashboard Layout Example;

```
----------------------------------------------------
                 Hospital Dashboard
----------------------------------------------------

[ Total Patients KPI ]

----------------------------------------------------
| Patient Distribution | Department Cost Analysis |
----------------------------------------------------

----------------------------------------------------
|        Outcome Distribution Pie Chart           |
----------------------------------------------------

Filters:
[Region ▼]
[Department ▼]

----------------------------------------------------
```

Dashboard Insights :

- Compare patient volume by region.
- Analyze department-wise treatment costs.
- Monitor patient outcomes.

- Interactively filter data.

Q9 Storytelling Flow Create a Tableau Story with minimum 4 Story Points:

1.Patient Distribution

2.Department Cost Analysis

3.Outcome Analysis

4.Doctor Performance.

SOL : Create a Tableau Story with Minimum 4 Story Points.

Steps;

1. Click **New Story**.
2. Set Story Size → Automatic.
3. Add Story Points one by one.

Story Point 1

Title;

Patient Distribution Analysis

Visualization;

Region Bar Chart

Insight;

Identifies regions with highest patient admissions.

Story Point 2

Title;

Department Cost Analysis

Visualization ;

Treatment Cost by Department

Insight;

Shows which departments consume maximum healthcare resources.

Story Point 3

Title ;

Outcome Analysis

Visualization;

Outcome Pie Chart

INSIGHT:

Evaluates patient recovery and critical cases.

Story Point 4

Title;

Doctor Performance Analysis

Visualization;

Top Doctors Ranked Bar Chart

Insight;

Highlights doctors handling high-value treatments.

Story Flow Layout;

```
Story Point 1
↓
Patient Distribution

Story Point 2
↓
Department Cost Analysis

Story Point 3
↓
Outcome Analysis

Story Point 4
↓
Doctor Performance
```

Q10 Based on your visuals, answer:

1.Which department generates the highest treatment cost?

2.Which region has more critical patients?

3.What business decisions can hospital management take from these insights?

Sol : 1. Which Department Generates the Highest Treatment Cost?

How to Find;

- Open Department Cost Chart.
- Identify the department with the tallest bar.

Example Answer;

If Cardiology has the highest value:

Cardiology generates the highest treatment cost because it handles complex procedures and expensive treatments.

(Replace with your actual department after checking the chart.)

2. Which Region Has More Critical Patients?

How to Find;

1. Filter Outcome = Critical.
2. Create Region vs Patient Count chart.
3. Identify the highest region.

Example Answer;

The North Region has the highest number of critical patients.

(Replace with your actual result.)

3. Business Decisions Hospital Management Can Take

Resource Allocation;

- Allocate more doctors and beds to high-patient regions.

Budget Planning;

- Increase budget for departments generating high treatment costs.

Improve Patient Outcomes;

- Analyze causes of critical cases.

- Improve treatment protocols.

Doctor Workload Management;

- Balance workload among doctors.
- Reward top-performing doctors.

Regional Healthcare Expansion;

- Expand facilities in regions with increasing patient demand.